

Precision Agriculture Sensor Notes

Project summary

This sample note describes sensor data sources for precision agriculture. The aim is to combine remote sensing, field measurements, and weather data to support crop monitoring and decision support.

Sensor and data sources

Relevant data sources include Sentinel-2 multispectral imagery, drone RGB imagery, drone multispectral imagery, soil moisture sensors, canopy temperature sensors, weather station data, yield maps, and farmer field observations. Sentinel-2 can support vegetation index extraction such as NDVI and red-edge indices. Drone data can provide higher spatial resolution for disease patches, crop gaps, and local stress symptoms.

Machine learning tasks

Typical machine learning tasks include crop stress classification, yield prediction, disease detection, weed detection, irrigation recommendation, and anomaly detection. Training labels may come from field scouting, yield maps, laboratory analysis, or expert annotation of image data.

Validation considerations

Validation should avoid leakage across fields and dates. A useful split is field-level or season-level separation, rather than random pixels from the same field. Metrics may include F1 score for stress classification, mean absolute error for yield prediction, and calibration metrics when uncertainty estimates are used.